

Discrete open loop boost converter and clock

This project involves the design, simulation, implementation, and experimental characterization of an open-loop DC–DC boost converter using discrete electronic components. The switching signal was generated using a two-transistor astable multivibrator followed by a CD40106 Schmitt Trigger inverter, eliminating the need for a microcontroller or dedicated PWM controller IC. The project aims to provide a practical understanding of boost converter operation, switching power electronics, component selection, and the influence of switching parameters on converter performance. Extensive hardware experimentation was carried out to investigate load regulation, transistor base-drive optimization, and the differences between theoretical predictions, LTspice simulation, and practical implementation.

Technical Rationale and Architectural Significance

- The project implements an open-loop DC–DC boost converter using discrete electronic components instead of employing dedicated boost converter ICs or microcontroller-based PWM generation.
- The switching signal is generated using a two-transistor astable multivibrator followed by a CD40106 Schmitt Trigger inverter, resulting in a completely hardware-based PWM generation circuit.
- The converter was first designed and analyzed in LTspice before being implemented in hardware, allowing comparison between theoretical predictions and practical performance.
- Experimental characterization was carried out under different loading conditions to study load regulation, output voltage variation, and power delivery capability.
- The influence of transistor base-drive resistance on converter performance was investigated experimentally, demonstrating the importance of proper transistor saturation in switching power converters.
- The switching frequency and duty cycle were measured, providing hardware verification of the designed oscillator.
- The project demonstrates the practical differences between ideal boost converter theory and real-world implementation, including the effects of transistor saturation, diode forward voltage, parasitic resistances, and switching losses.

System Architecture

The boost converter system is organized into four primary functional blocks: a PWM generation unit, a switching stage, an energy storage stage, and an output filtering stage. Together, these blocks convert a low-voltage DC input into a higher DC output voltage by periodically storing

energy in an inductor and subsequently transferring it to the output capacitor. The overall system architecture is illustrated in **Figure 1**.

The PWM generation unit consists of a discrete two-transistor astable multivibrator followed by a CD40106 Schmitt Trigger inverter. The astable multivibrator generates a continuous square-wave signal, while the Schmitt Trigger improves waveform quality by producing clean switching pulses with sharp rise and fall times. The resulting signal is used to drive the switching transistor. **Figure 2** shows the output of PWM stage obtained in simulation.

The switching stage consists of a 2N2222A NPN transistor operating as an electronic switch. During the ON interval of the switching signal, the transistor saturates and allows current to flow through the inductor, storing energy in its magnetic field. When the transistor switches OFF, the stored magnetic energy is released.

The energy transfer stage comprises a 100 μ H inductor and a 1N5819 Schottky diode. During the OFF interval, the inductor forces current through the forward-biased Schottky diode into the output capacitor and load. The Schottky diode prevents reverse current flow from the output capacitor back into the switching stage during the transistor ON interval.

The output filtering stage consists of a 220 μ F electrolytic capacitor connected across the load. The capacitor stores energy delivered by the inductor and supplies current to the load between switching cycles, thereby reducing output voltage ripple and maintaining a relatively constant DC output voltage.

The converter is powered from a single 1.5 V AA cell and operates in an open-loop configuration. Unlike closed-loop boost converters, no feedback mechanism is employed to regulate the output voltage. Consequently, the output voltage varies with changes in load resistance and the practical characteristics of the switching components.

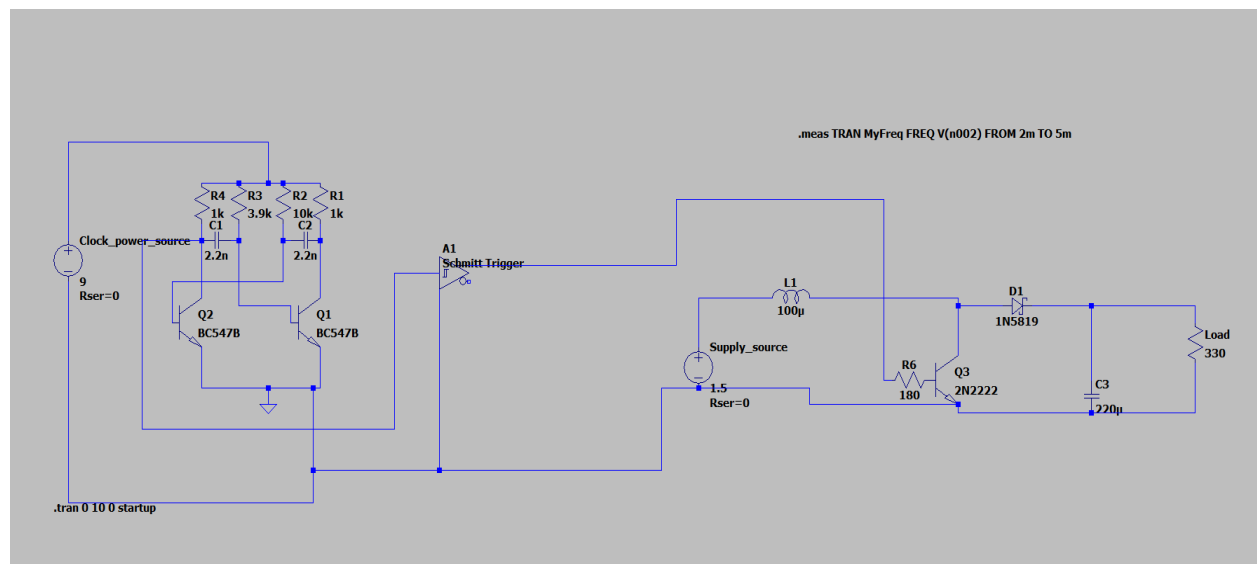


Figure 1

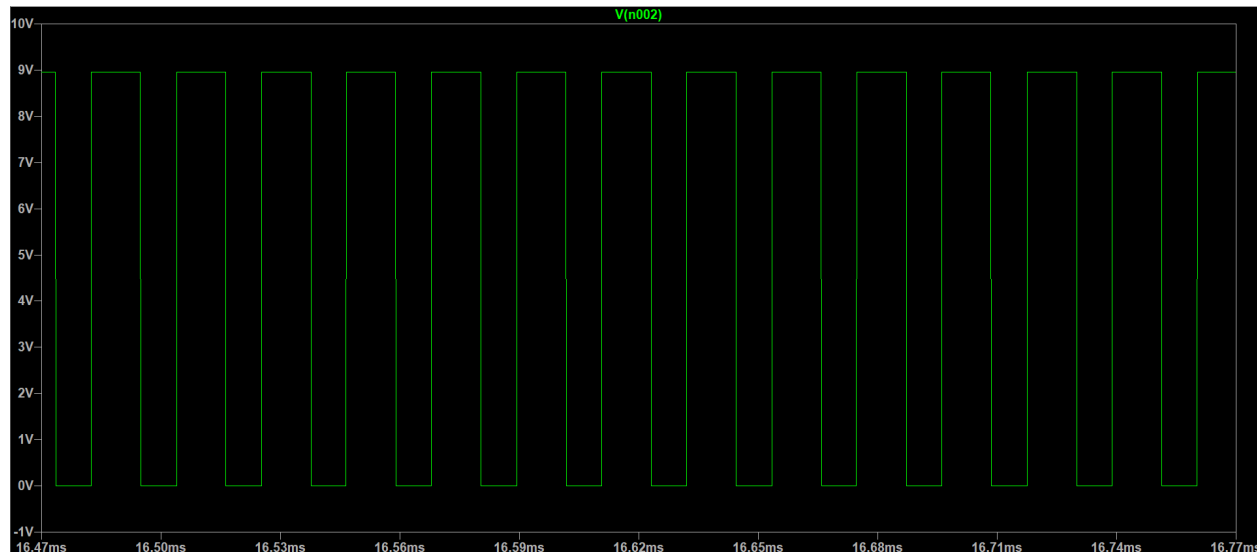


Figure 2

Key Design Decisions

A switching frequency of approximately 50kHz was selected for the converter, however due to unavailability of exact values of components and tolerances it shifted to 44.5kHz. This frequency provided a suitable compromise between converter performance and transistor switching characteristics. Preliminary experimentation at higher frequencies resulted in incomplete transistor turn-off due to charge storage effects, whereas significantly lower frequencies reduced the energy transfer rate and increased output voltage ripple.

The switching waveform was generated using a discrete **two-transistor astable multivibrator** followed by a **CD40106 Schmitt Trigger** inverter instead of a microcontroller. This approach minimized system complexity, eliminated software dependencies, and provided a dedicated hardware PWM source suitable for studying the converter's operating principles.

A **2N2222 transistor** was selected as the switching device because of its availability and its capability to operate at the chosen switching frequency. Although BJTs require continuous base current and exhibit higher switching losses than MOSFETs, they provide an effective platform for understanding transistor-based switching converters and are cost effective.

A **100 μ H inductor** was selected to provide sufficient energy storage while maintaining acceptable current levels at the selected switching frequency. The inductor value was also compatible with the desired operating frequency and the current handling capability of the switching transistor.

A **1N5819 Schottky** diode was employed because of its low forward voltage drop and fast reverse recovery characteristics. Compared with conventional PN junction diodes, the Schottky

diode reduces conduction losses and improves converter efficiency, particularly in low-voltage applications.

A **220 μF electrolytic capacitor** was selected as the output filter capacitor to reduce output voltage ripple and maintain a stable DC output under varying load conditions. Although ceramic capacitors possess lower equivalent series resistance (ESR), the larger capacitance of the electrolytic capacitor provided superior energy storage for this prototype.

During experimental testing, the transistor base-drive resistance was systematically optimized. Initial values of 2.2 k Ω and 1 k Ω resulted in poor load regulation due to insufficient transistor saturation. Progressive reduction of the base resistance improved converter performance, and a value of 180 Ω was found to provide the best compromise between output voltage and switching performance. Further reduction to 90 Ω produced only marginal improvement, indicating that the converter performance had become limited by other practical non-idealities rather than base-drive strength alone.

Practical Challenges and Debugging

The initial hardware implementation did not produce the expected output voltage predicted by LTspice simulation. Although the converter generated approximately 6.9 V under no-load conditions, the output voltage dropped significantly when a load was connected, indicating that the converter was unable to transfer sufficient power to the output.

To identify the cause of this behavior, the switching frequency and duty cycle were experimentally verified using the ESP32 microcontroller. The measured frequency was approximately **44.45 kHz** with a duty cycle of **82.3%**, confirming that the PWM generation stage was operating as designed. This eliminated the oscillator as the primary source of the performance degradation.

Further investigation focused on the transistor switching stage. It was observed that increasing the transistor base-drive current substantially improved the converter's loaded output voltage. A systematic study was therefore carried out by varying the transistor base resistor. Reducing the base resistance from **2.2 k Ω** to **180 Ω** increased the output voltage across a **330 Ω** load from approximately **2.0 V** to **4.4 V**, demonstrating the importance of adequate transistor saturation in low-voltage switching converters. Further reduction of the base resistance to **90 Ω** produced only **marginal improvement**, indicating that the converter performance had approached the practical limits imposed by the selected components.

The practical output voltage was also observed to be lower than the theoretical prediction. This difference is attributed to several non-idealities neglected in the ideal boost converter model, including transistor saturation voltage, Schottky diode forward voltage, inductor winding resistance, battery internal resistance, switching losses, and parasitic resistances associated with breadboard implementation.

The debugging process emphasized the importance of systematic hardware characterization. Rather than relying solely on theoretical analysis or simulation, practical measurements of switching frequency, duty cycle, base-drive conditions, and load regulation were used to identify performance limitations and guide subsequent design improvements.

Experimental Measurements and Results

The completed boost converter was experimentally characterized to evaluate its practical performance under different operating conditions. The switching characteristics were first measured to verify the operation of the PWM generation stage, followed by load regulation tests and an investigation of the influence of transistor base-drive resistance on converter performance.

Switching Characteristics

The switching frequency and duty cycle of the converter were measured using the ESP32 MCPWM Capture peripheral. The measured values are summarized in Table 1.

Parameter	Measured Value
Input Voltage	1.5V
Switching Frequency	44.45kHz
Duty Cycle	82.3%

Table 1: Measured Switching Parameters

The measured switching frequency closely matched the intended design value, confirming proper operation of the astable multivibrator and Schmitt Trigger driver stage.

Output Voltage under Different Loads

The converter output voltage was measured for various load resistances while maintaining the same switching frequency and duty cycle.

Load Resistance	Output Voltage
No load	7.8V
2.2k Ω	7.0V
1k Ω	6.4V
470 Ω	5.1V
330 Ω	4.4V
180 Ω	3.0V
90 Ω	2.0V

Table 2: Load Regulation

As expected, the output voltage decreased with decreasing load resistance due to increased output current demand and practical losses within the converter.

Output Power

The corresponding output current and output power were calculated from the measured output voltages.

Load Resistance	Output Current	Output Power
2.2k Ω	3.18mA	22.3mW
1k Ω	6.40mA	41.0mW
470 Ω	10.85mA	55.3mW
330 Ω	13.33mA	58.7mW
180 Ω	16.9mA	52.4mW
90 Ω	22.2mA	44.4mW

Table 3: Output Power

The converter delivered its maximum measured output power at approximately the 330 Ω load, beyond which further reduction in load resistance resulted in reduced output voltage and output power.

Effect of Base Drive Resistance

To investigate the influence of transistor saturation on converter performance, the transistor base resistance was varied while maintaining a constant load resistance of 330 Ω .

Base Resistance	Output Voltage
2.2k Ω	2.0V
1k Ω	3.0V
470 Ω	3.8V
180 Ω	4.4V
90 Ω	4.5V

Table 4: Effect of Base Resistance

The experimental results indicate that reducing the base resistance improved transistor saturation and significantly enhanced converter performance under load. However, reducing the base resistance below 180 Ω produced only marginal improvement, suggesting that the converter performance had become limited by other practical factors.

Comparison with Theoretical Prediction

For the measured duty cycle of **82.3%**, the ideal boost converter equation predicts

$$V_{out} = V_{in}/(1-D)$$

Using an input voltage of **1.5 V**, the theoretical no-load output voltage is approximately **8.5 V**. The experimentally measured no-load output voltage was **7.8 V**, corresponding to an error of approximately **8%**. This difference is primarily attributed to transistor saturation voltage, diode forward voltage, winding resistance, switching losses, and other practical non-idealities neglected in the ideal boost converter model.

Base Drive Current and Base Drive Power

The transistor base-drive current was estimated by measuring the voltage across the **180 Ω** base resistor while the converter was operating under load. A voltage of **2.2 V** was measured across the resistor.

Using Ohm's law, we get base current as **12.2mA**, and hence power dissipated across resistor is **26.9mW**

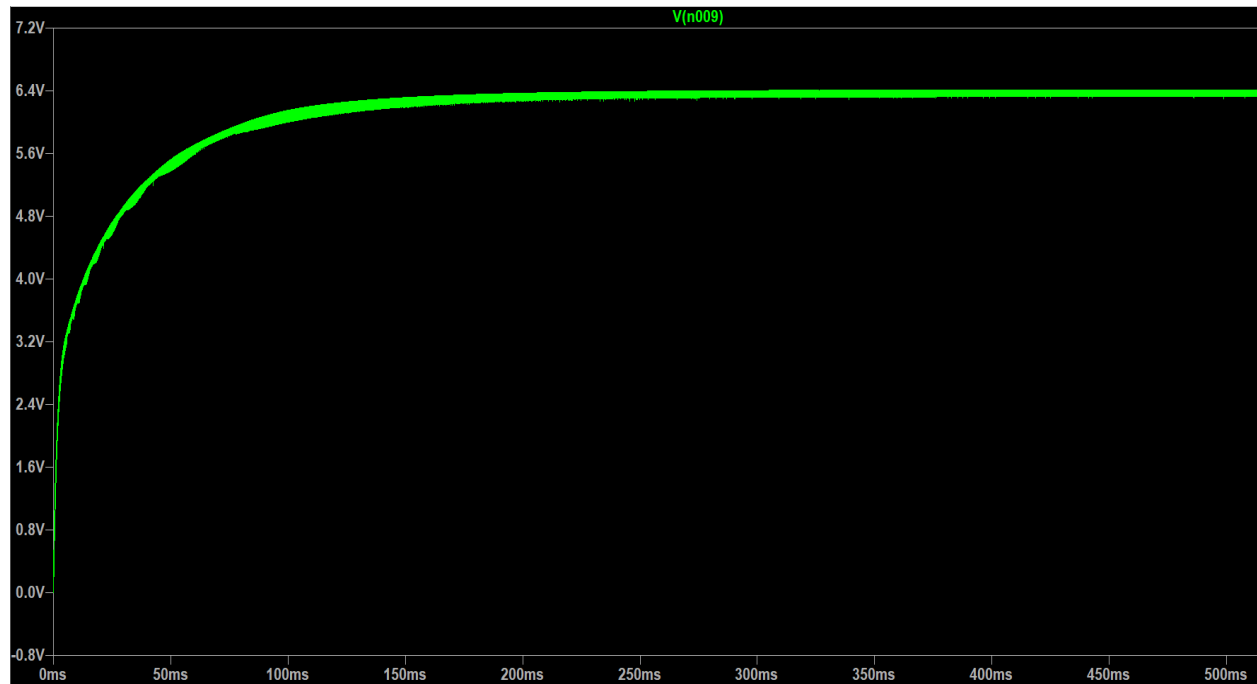
Assuming a base-emitter voltage of approximately 0.7 V, the power dissipated across the base-emitter junction is $0.7 \times 12.2\text{mA} = \mathbf{8.5mW}$

So total base drive power is $8.5\text{mW} + 26.9\text{mW} = \mathbf{35.4mW}$

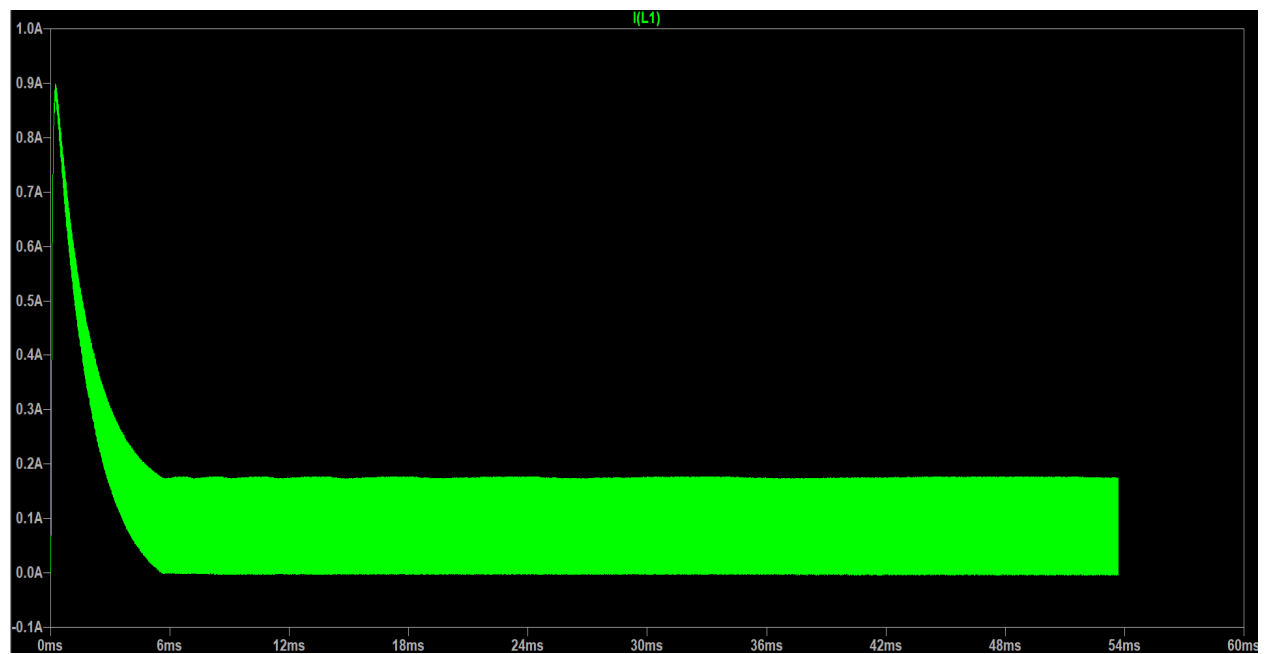
What this Project Demonstrates

- Practical understanding of the operating principles of DC–DC boost converters and switching power electronics.
- Ability to design, simulate, and analyze power electronic circuits using LTspice before hardware implementation.
- Understanding of transistor operation in switching applications, including transistor saturation, base-drive requirements, and switching losses.
- Practical knowledge of passive component selection, including inductors, capacitors, and Schottky diodes, based on circuit requirements.
- Experience in designing hardware PWM generation circuits using a discrete astable multivibrator and Schmitt Trigger inverter.
- Experimental characterization of converter performance through systematic measurements of switching frequency, duty cycle, load regulation, and output power.
- Hardware debugging and performance optimization through controlled experimentation, including investigation of transistor base-drive resistance and comparison between theoretical predictions and practical observations.
- Appreciation of the practical differences between ideal converter theory, circuit simulation, and real-world hardware implementation.
- Understanding of engineering trade-offs involving efficiency, switching speed, transistor drive requirements, and component limitations in power electronic systems.

Circuit Pictures and Simulation output

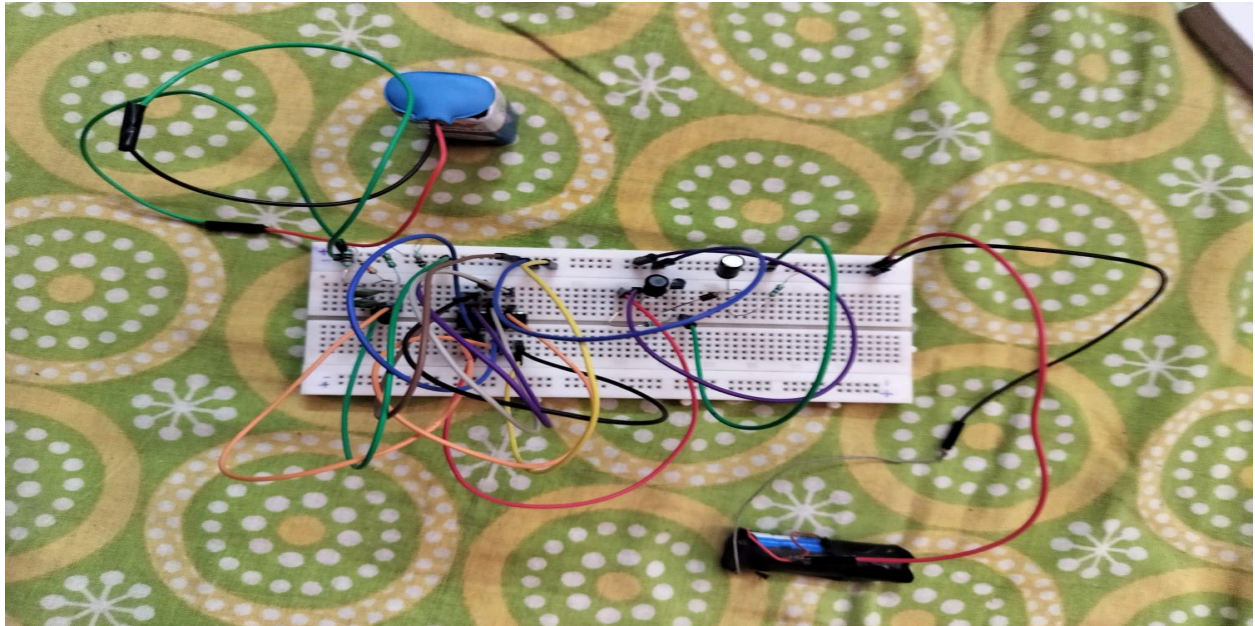


Simulated output across load resistor of 470Ω



Current across inductor for the above output

Note :- Initial current surge in inductor is of very less duration it won't harm the inductor, the reason of it occurring is possibly the initial empty state of the circuit, the empty inductor absorbs and dumps a lot of energy in empty capacitor, but soon the current settles down.



Circuit picture

Limitations and Future Improvements

Limitations

- The converter operates in an **open-loop configuration** and therefore does not regulate its output voltage. The output voltage varies with changes in load resistance and input supply voltage.
- The use of a **2N2222 BJT** as the switching device requires continuous base-drive current, resulting in higher switching losses and lower efficiency compared to MOSFET-based converters.
- The converter was implemented on a **breadboard**, making it susceptible to parasitic inductance, contact resistance, and mechanical disturbances, all of which influence the practical performance.
- The practical output voltage and power are lower than the theoretical values due to Schottky diode forward voltage, inductor winding resistance, battery internal resistance, and other non-ideal component characteristics.

- The converter is designed as a **low-power proof-of-concept prototype** and is not intended for high-power applications without significant redesign of the power stage.

Future Improvements

- Implement **closed-loop voltage regulation** using a microcontroller or dedicated PWM controller to maintain a constant output voltage under varying load conditions.
- Replace the BJT switching transistor with a **logic-level MOSFET** to reduce switching losses, eliminate continuous base-drive power consumption, and improve overall converter efficiency.
- Design and fabricate a **Printed Circuit Board (PCB)** to minimize parasitic effects, improve reliability, and enhance switching performance.
- Investigate the influence of switching frequency, inductor value, and output capacitance on converter efficiency and load regulation through systematic experimental characterization.
- Incorporate **current limiting, over-current protection, and soft-start circuitry** to improve reliability and prevent excessive startup currents.
- Extend the design to support **higher output power** by selecting power-rated switching devices, inductors with higher saturation current, low-ESR capacitors, and appropriate thermal management techniques.

Conclusion

An open-loop DC–DC boost converter was successfully designed, simulated, implemented, and experimentally characterized using discrete electronic components. The hardware prototype demonstrated the ability to step up a 1.5 V input to a maximum no-load output voltage of approximately 7.8 V while operating at a switching frequency of 44.45 kHz. Experimental measurements closely followed theoretical predictions under no-load conditions and highlighted the influence of practical non-idealities under load. The project provided valuable insight into switching power electronics, component selection, hardware debugging, and the differences between ideal analysis and real-world implementation. Overall, the work successfully achieved its objective of demonstrating the practical operation and characterization of a discrete boost converter.

